System Analysis • UML • Project Report

EduZone

Teresa Yuardy | 2026

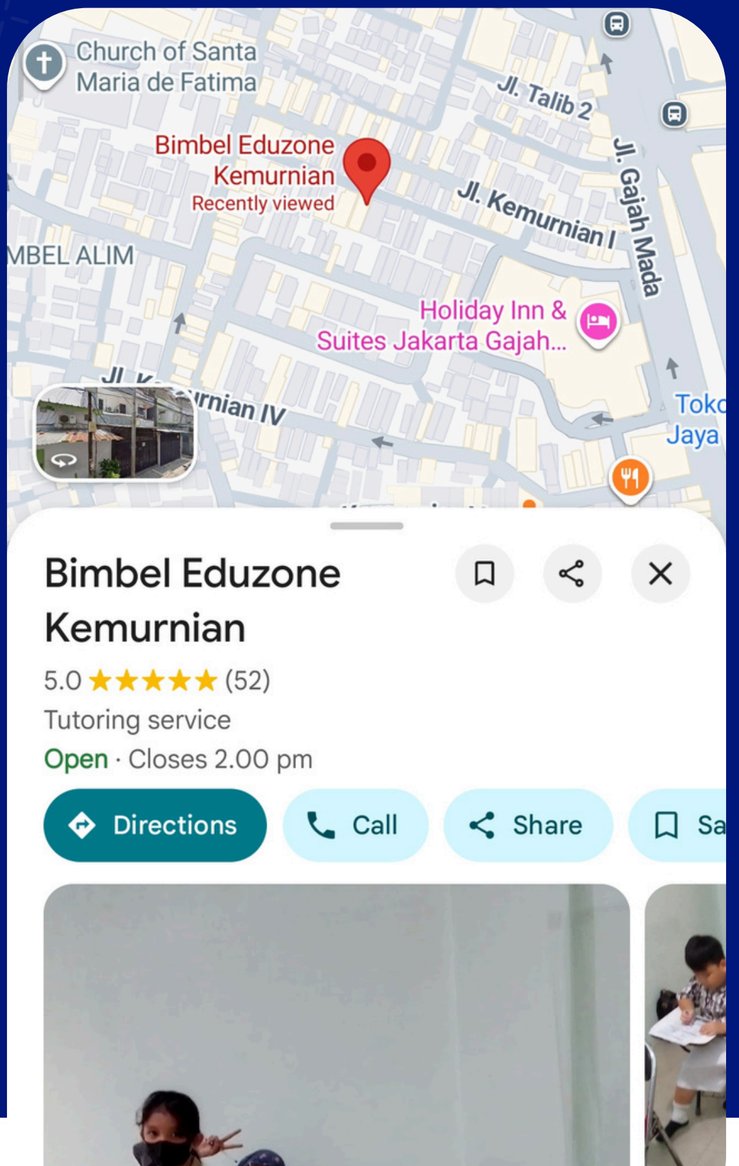
Academic Project

Software Development · 4th Semester · Final Examination Project

About EduZone

- **EduZone** is a web-based platform designed to support offline tutoring activities by providing students, tutors, and administrators with an integrated system for managing academic and administrative information.
  Rather than replacing face-to-face learning, EduZone serves as a complementary tool that helps make tutoring activities more **organized**, **efficient**, and **accessible**.

EduZone's location



Users and System Features

Administrator

- Add students & tutors
- Manage payments
- Manage schedules
- Monitor attendance

Tutor

- Record attendance
- Upload materials
- Upload excercises
- View teaching schedule

Student

- Check attendance
- View materials
- Complete exercise
- Check payment

8 System Models

- Context Diagram
- Data Flow Diagram
- Use Case Diagram
- Activity Diagram
- Class Diagram
- Entity Relationship Diagram
- Sequence Diagram
- Communication Diagram

My Contribution

In this project, I was responsible for the **system analysis** and **final project report**, while the web implementation was handled by my teammate.

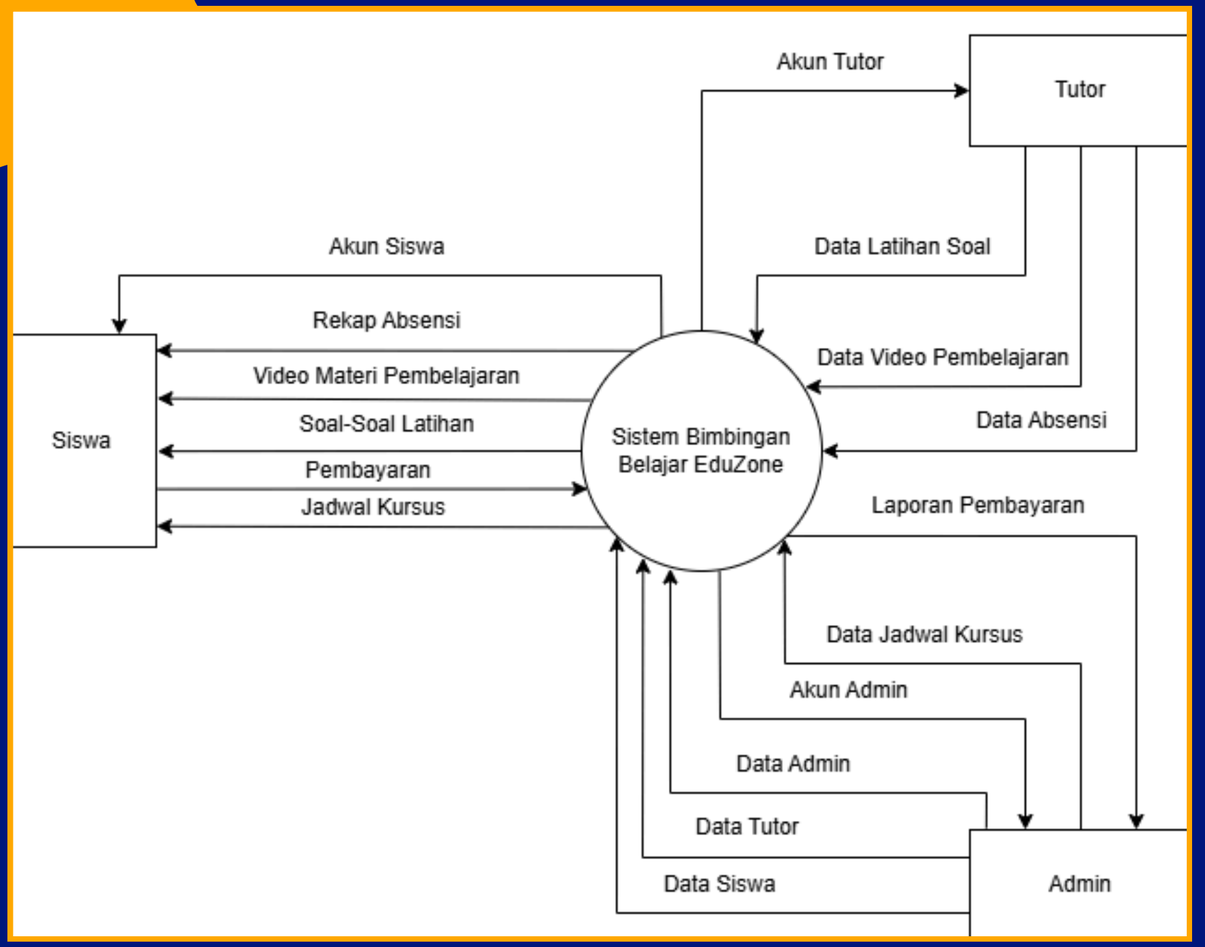
INFORMATION FLOW

The system's information flow was modeled to illustrate how data moves between users, processes, and the EduZone system.



Context Diagram

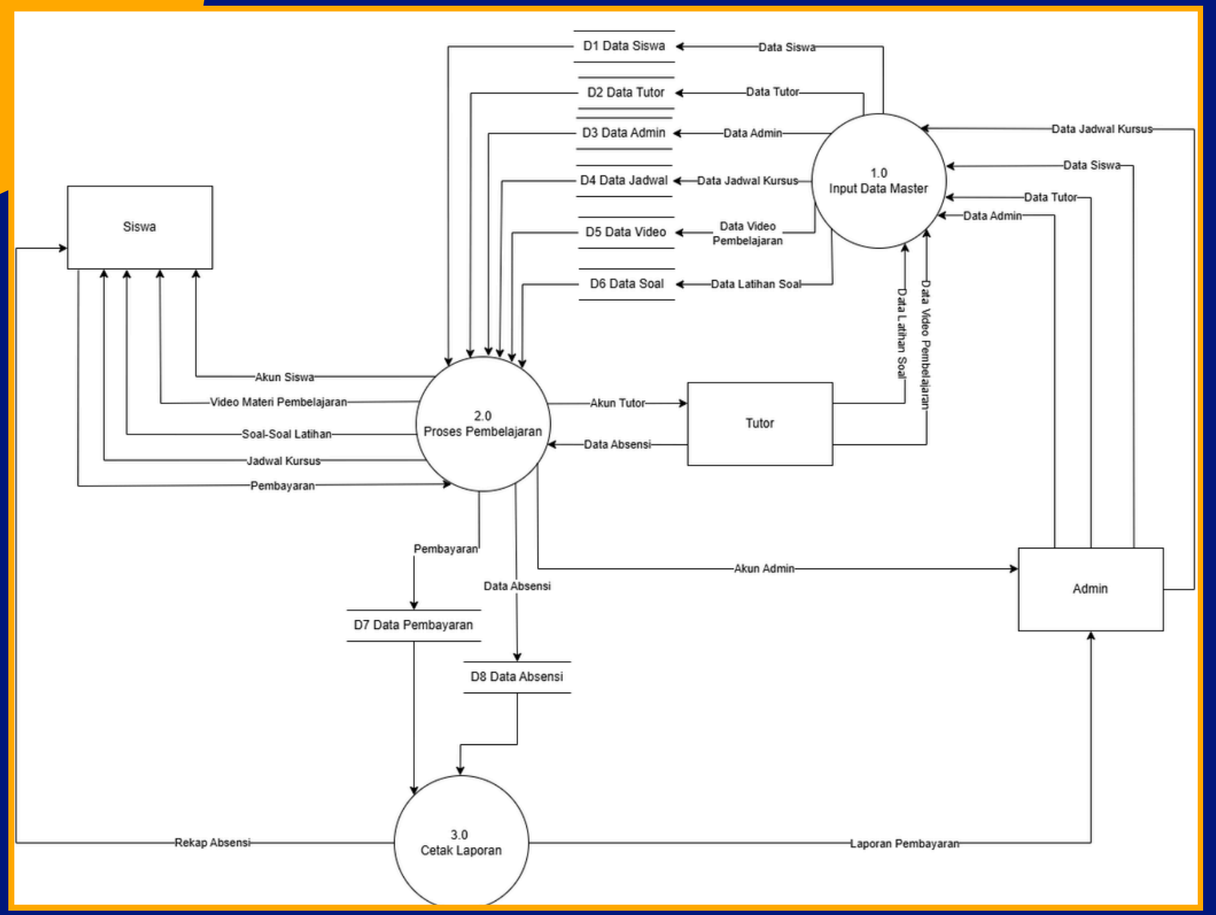
- The **Context Diagram** provides a high-level overview of EduZone by showing its interactions with the three main actors: **Admin**, **Tutor**, and **Student**.



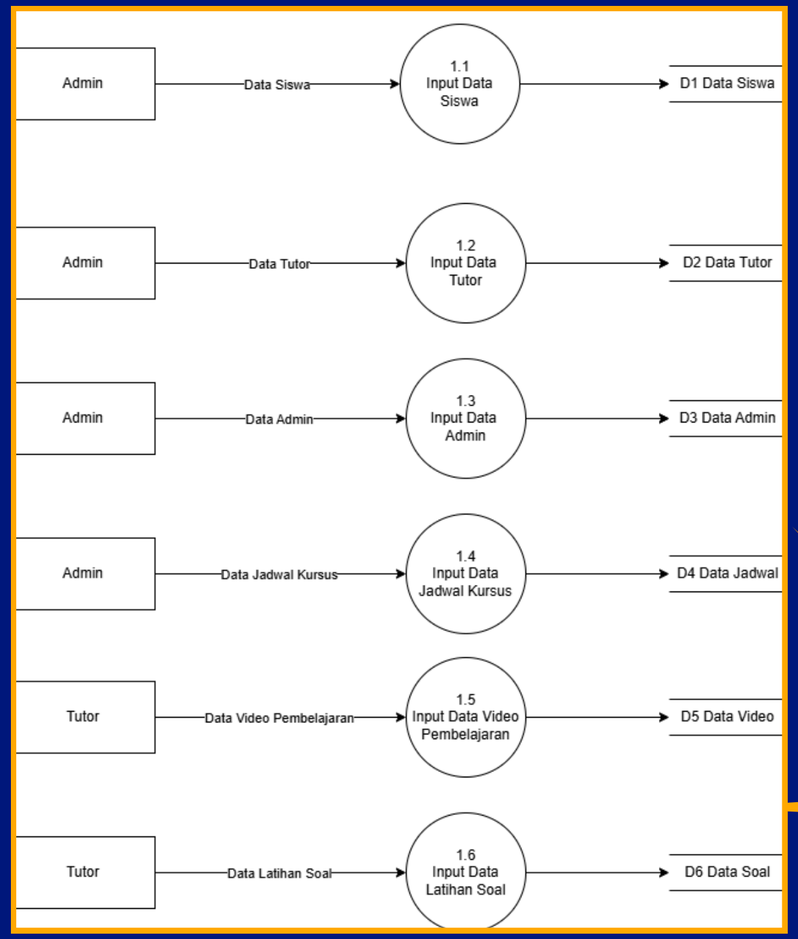
Data Flow Diagram

- The **Data Flow Diagram** decomposes EduZone into its main processes and illustrates how information flows between users, processes, and data stores.

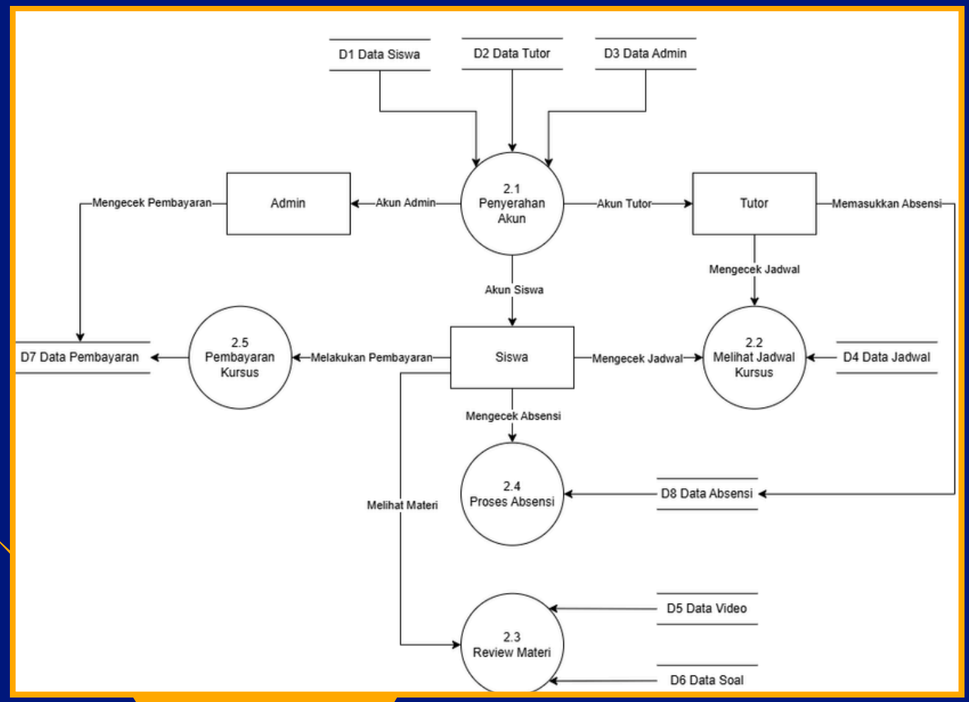
DFD Level 0



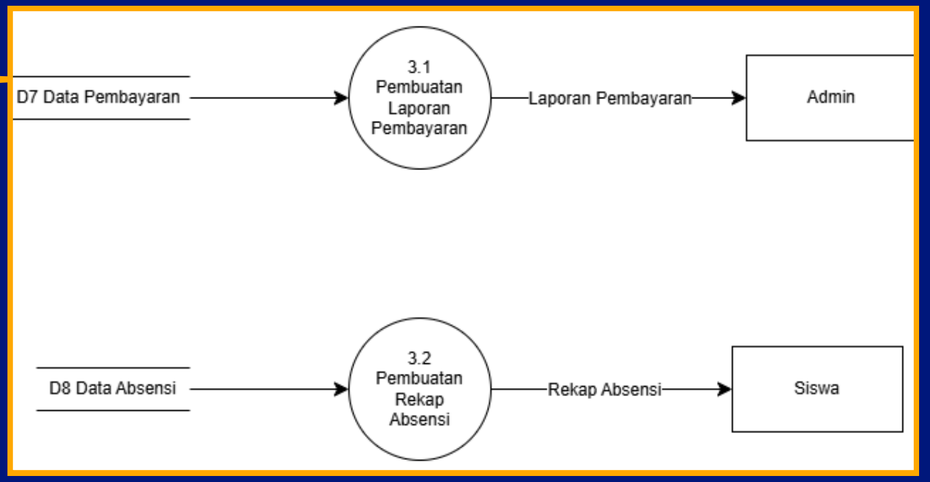
DFD Level 1 Process 1



DFD Level 1 Process 2



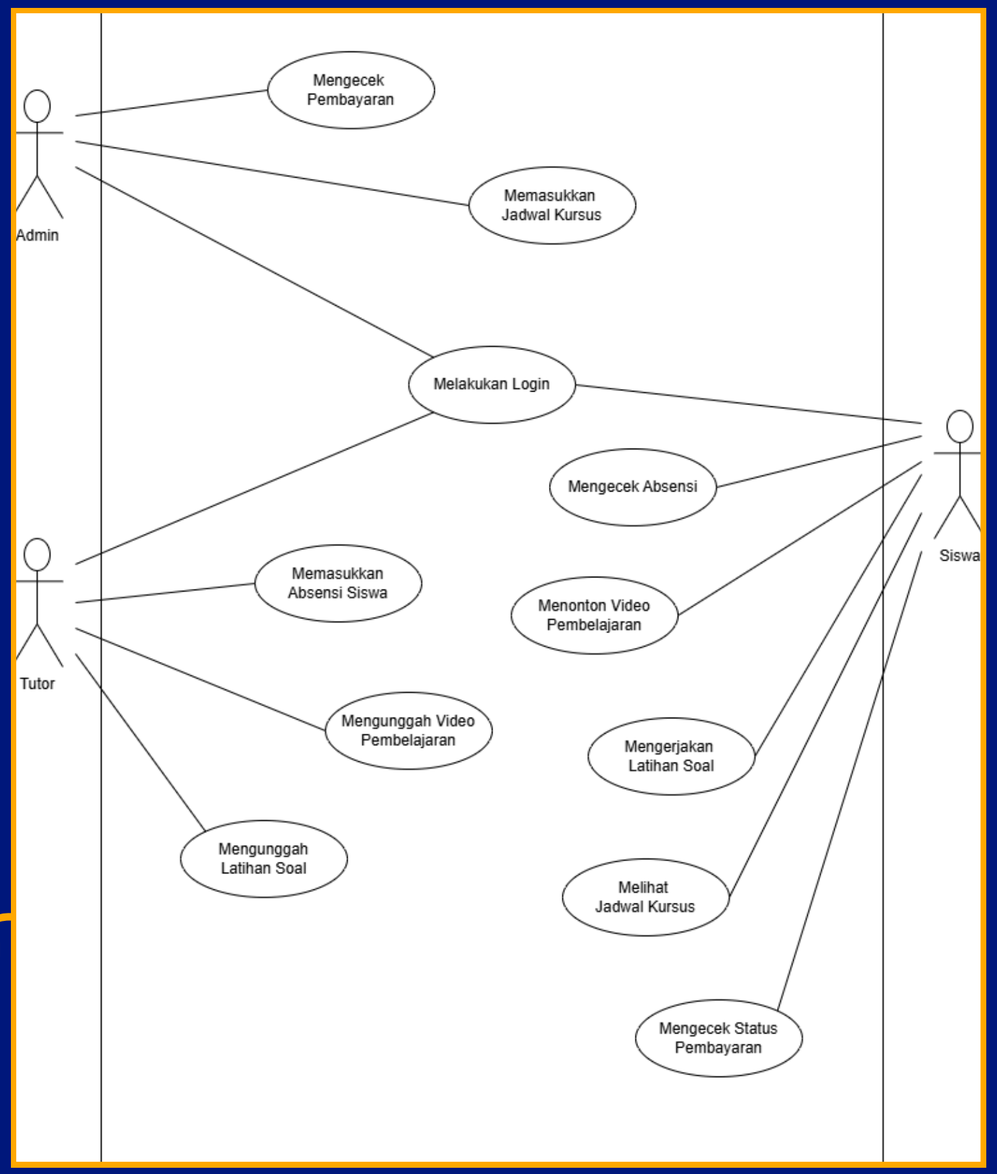
DFD Level 1 Process 3



PROCESS MODELING

Process modeling describes how users interact with EduZone and how system processes are carried out to fulfill each user's needs.

Use Case Diagram → **Activity Diagram**



Use Case Diagram

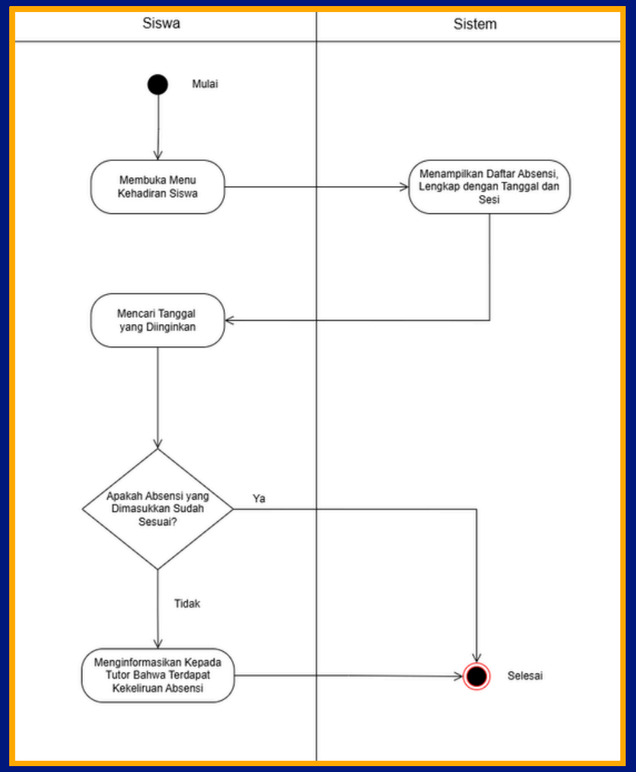
- The **Use Case Diagram** illustrates the interactions between Admin, Tutor, Student, and the functionalities provided by the EduZone system.

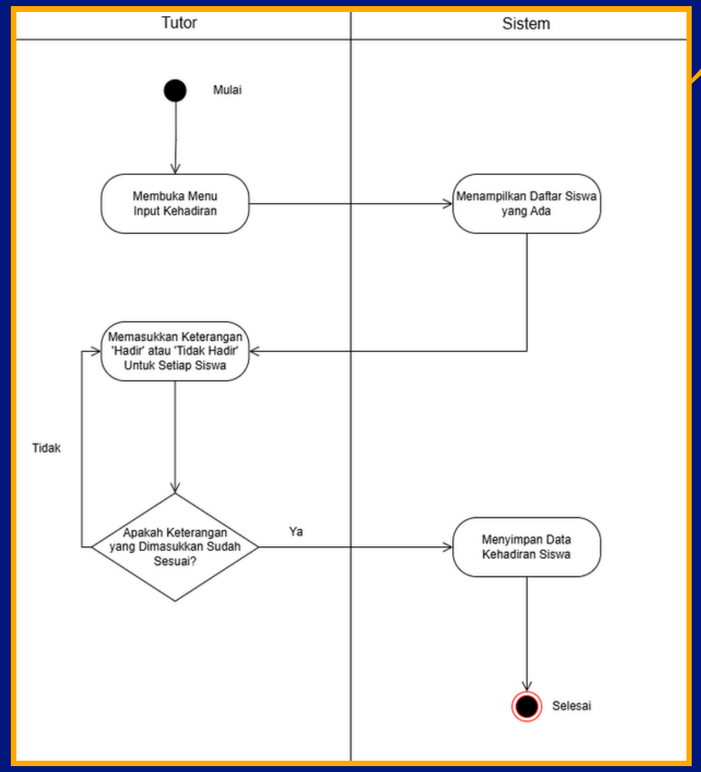
Activity Diagram

The **Activity Diagram** illustrates the workflow of selected processes in EduZone by showing the sequence of user actions and system responses.

Selected Process: Tutor — Record Attendance

Selected Process: Student — Check Attendance





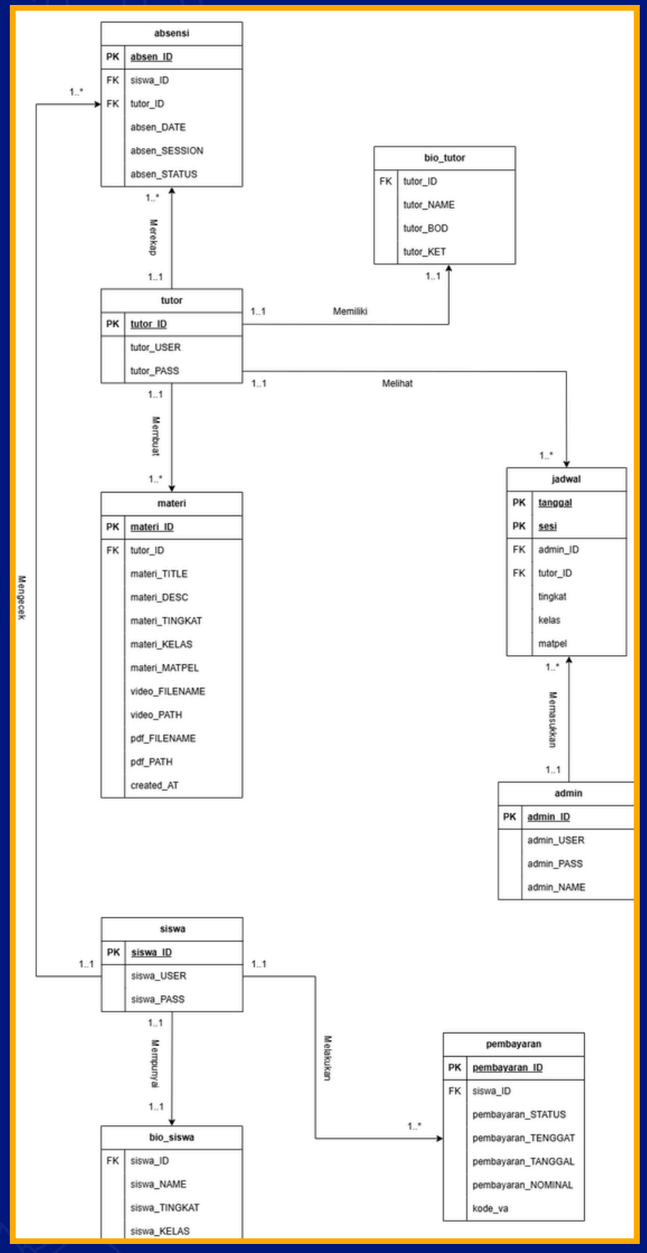
STRUCTURAL MODELING

Structural models describe how EduZone is organized through its database entities, classes, attributes, and relationships.

Class Diagram → **Entity Relationship Diagram**

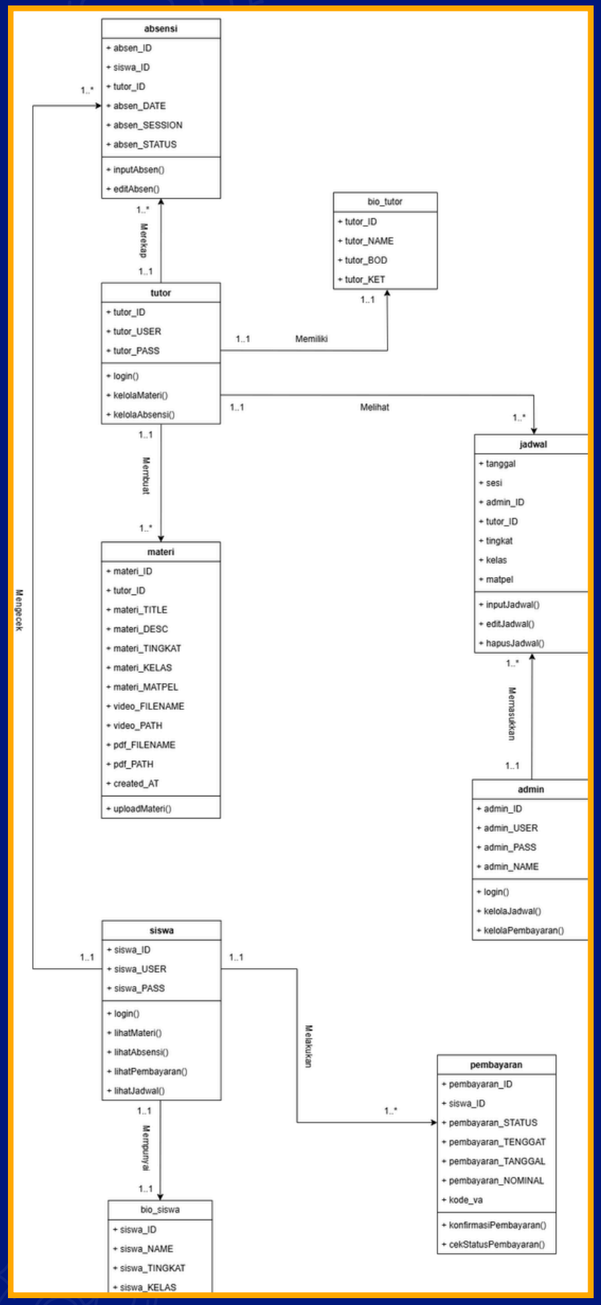
Entity Relationship Diagram

- The **Entity Relationship Diagram** defines the database structure of EduZone and illustrates the relationships between the entities involved in the system.



Class Diagram

- The **Class Diagram** represents the structural design of EduZone by defining its classes, attributes, methods, and relationships.



BEHAVIOUR MODELING

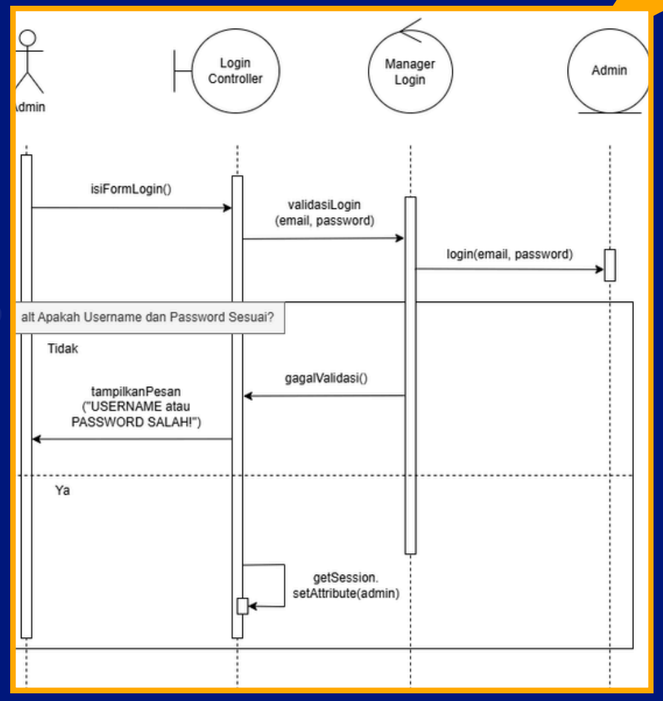
Behavioral models illustrate how users and system components interact with each other during specific system processes.

Sequence Diagram → **Communication Diagram**

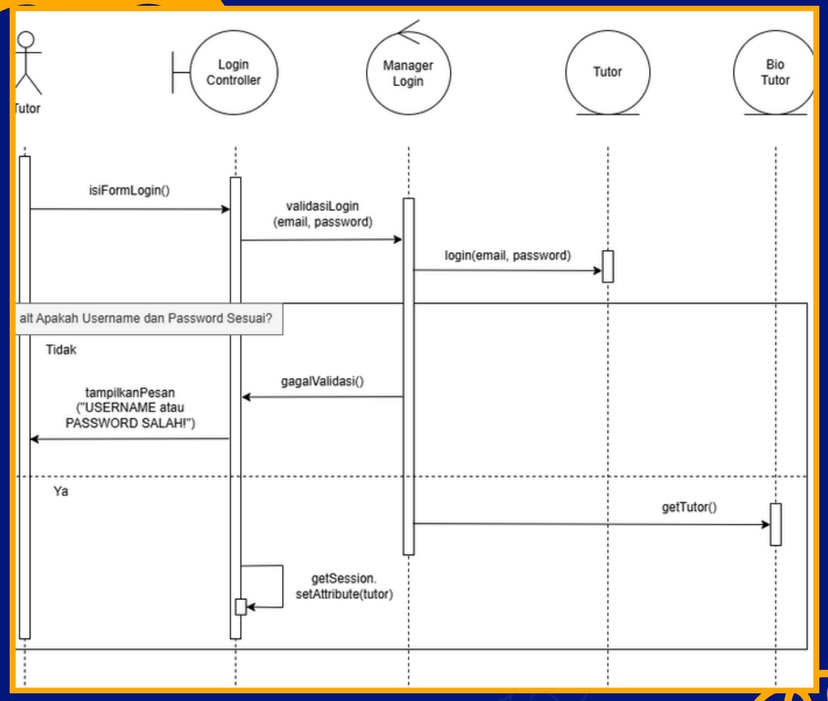
Sequence Diagram

The **Sequence Diagram** illustrates the sequence of interactions between users and system components during a specific process in EduZone.
Selected Process: User Login — Admin, Tutor & Student

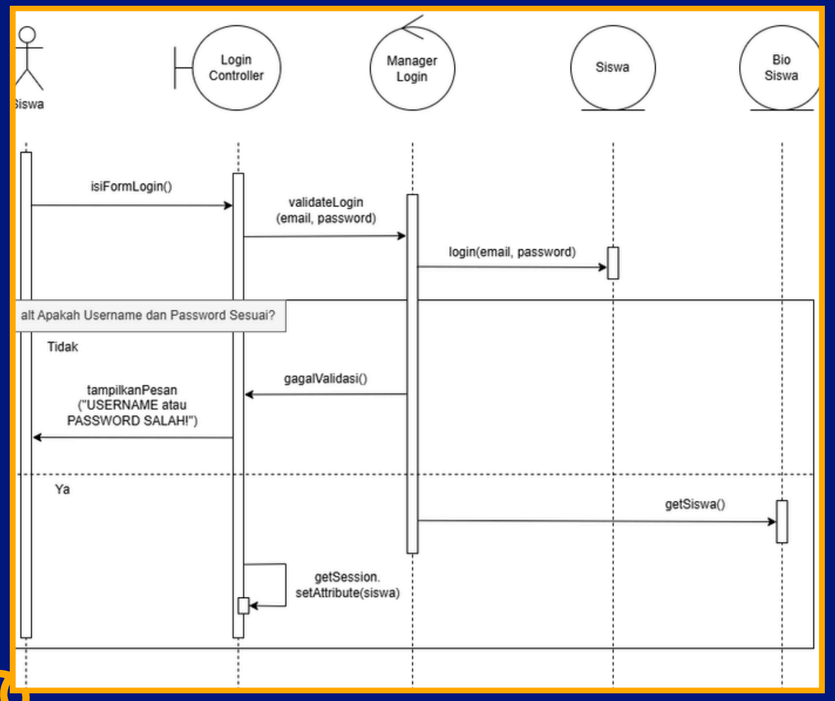
Administrator



Tutor



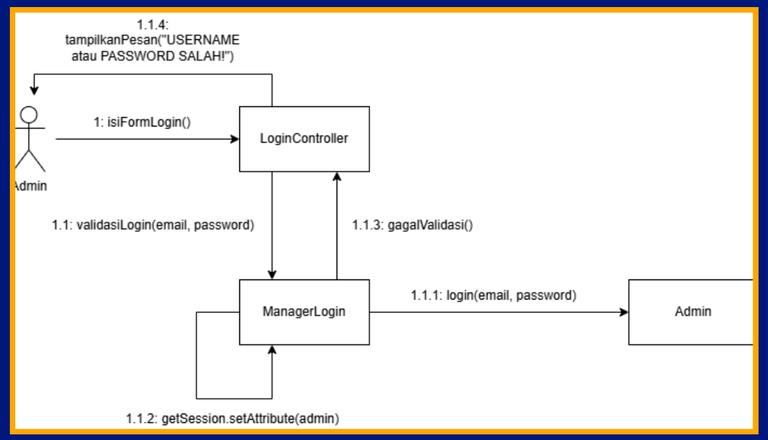
Student



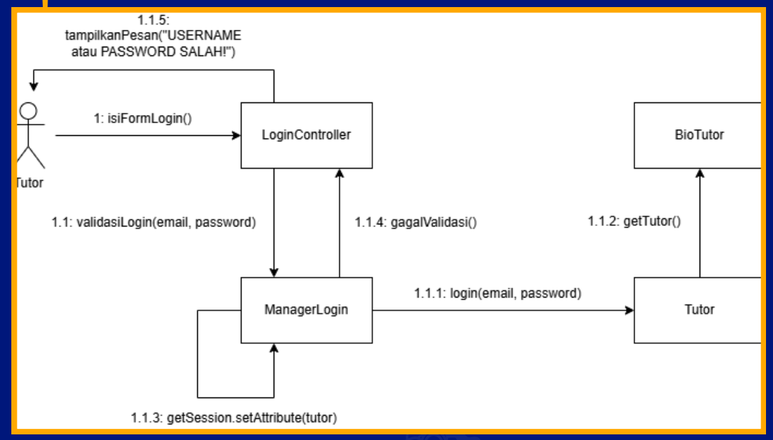
Communication Diagram

The **Communication Diagram** illustrates how users and system components communicate with each other to complete a specific process in EduZone.
Selected Process: User Login — Admin, Tutor & Student

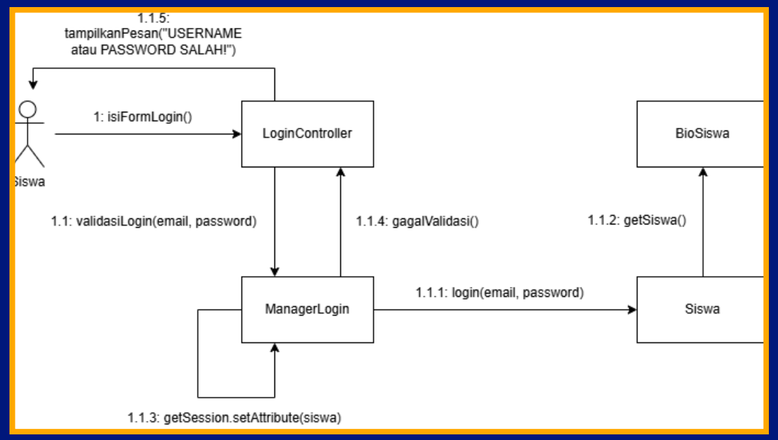
Administrator



Tutor



Student



Thank You for Exploring EduZone

From requirements analysis to system modeling and documentation.

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